Working Backwards Document: The Challenge of Cancellations

**Future Customer Quotes**

"We were tired of dealing with the constant headaches of staffing cancellations in the past few months. SF's new emphasis on successful staffing and making sure a replacement is available has been a game-changer for our business. We now have the peace of mind knowing that our workforce will be reliable and dependable."

- Olivia K., Admin

"SF used to leave me disappointed due to staffing issues and unreliable placements. But when I heard that they had made improvements to rectify those problems, I decided to give it another chance. I'm pleasantly surprised to find that the changes have resulted in successful staffing, and I'm back to using SF with renewed confidence."

- Thomas H., Admin

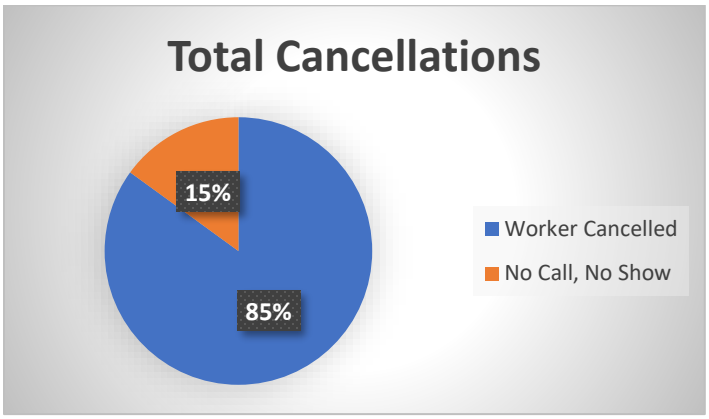
"Gone are the days of endlessly searching for available shifts. The improvements to SF's app have made it a breeze to find shifts that align with my preferences and expertise. The app's improved availability of shifts has made it an invaluable tool in my nursing career."

- Rachel D., Registered Nurse

**Internal FAQs**

1. **What is the problem?**

SF offers healthcare professionals (HCPs) the ability to select and secure shifts, a significant part of its value proposition. However, this feature presents challenges as HCPs are not always guaranteed to fulfill their commitments. This creates issues for both healthcare facilities (HCFs) and HCPs. HCFs rely on accurate staffing for each shift, and unexpected absences or no-shows by HCPs cause disruptions, these no-shows accounting for approximately 15% of all cancellations. Additionally, HCPs experience a reduction in available shifts when these no-shows "lock down" specific shifts, preventing them from booking the shifts that they would have otherwise worked.



2. **How do we solve it?**

To effectively address last-minute cancellations or no-shows, ensuring sufficient staffing for scheduled shifts is crucial. It has been noted that an attendance policy does not work to satisfy these cancellation concerns. We therefore examine how to ensure that there is ALWAYS someone available to take a shift, regardless of a cancellation occurring. Therefore, we will implement an advanced overfilling function that strategically assigns multiple healthcare professionals (HCPs) to a single shift. This approach safeguards SF and healthcare facilities (HCFs) from disruptions caused by individual cancellations. It's important to note that our overfilling solution goes beyond basic overbooking and involves a more sophisticated approach, which will be elaborated upon in the following sections.

3. **Why solve it?**

Addressing the staffing issue aligns directly with SF's long-term strategic goals of becoming a leader in the healthcare staffing industry. By prioritizing successful staffing and implementing solutions like the overfilling function, we establish ourselves as a reliable and innovative partner for healthcare facilities and professionals. This aligns with our vision of revolutionizing healthcare staffing and positions us for long-term success, market dominance, and the opportunity to shape the future of the industry.

4. **What are we classifying as success?**

In healthcare facility (HCF) settings, it is common to have multiple open shifts within a specific timeframe that require the same job role qualifications. To streamline this process, we categorize these shifts as "Shift Groups" and match them based on the aforementioned criteria of job role, time, and location.

True success lies in this effort is when facilities are able to have their shifts fulfilled at a 95% (or greater) success rate. This allows for HCF to be confident in SF that when they post a shift, that the shift will be filled AND that there will be minimal risk of a last minute cancellation.

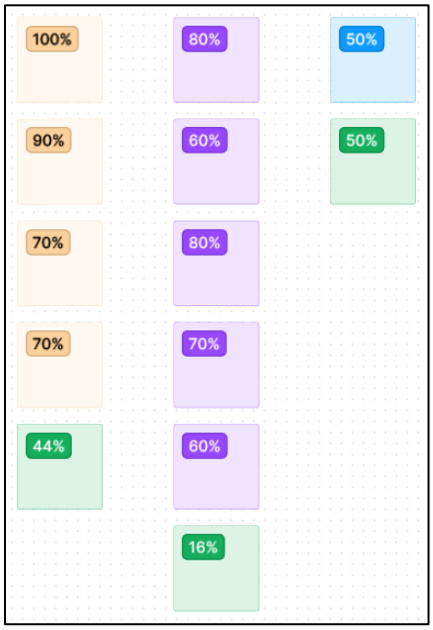
5. **What metric do we most want to move?**

The measure of success is determined by achieving a substantial improvement in the percentage of verified shifts compared to non-verified shifts, indicating shifts that have been confirmed and executed as scheduled. Presently, in the Cleveland region, the confirmed worked shifts for non-deleted shifts stand at a relatively low 40%. An effective initial rollout should aim to increase this metric by approximately 30%, resulting in a target of around 52% for the percentage of confirmed worked shifts. In practical terms, this signifies an additional 5000 shifts in Cleveland that are successfully confirmed and fulfilled, enhancing operational reliability and meeting staffing needs.

6. **What is being done to book multiple individuals into shifts?**

Within the framework of Shift Groups, multiple available shifts are presented to individuals. The objective of the new system is to strategically overfill these Shift Groups based on the likelihood that all individuals will fulfill their claimed shift. Leveraging our existing system that enforces

attendance rules and consequences, we can calculate the probability of an individual showing up for their shift using historical data (Shifts Worked/Total Shifts Booked). For newly onboarded healthcare professionals (HCPs), a preliminary probability of 90% is set until sufficient data is collected to confirm the average likelihood of individuals appearing for shifts in a specific market.




In the above image, the Shift Groups are collected by color with the respective percentage likelihood that the HCP is going to work that shift. The green colored boxes represent the probability that all of those individuals are going to work their shift (i.e., what are the odds that someone cancels). This is calculated by simple multiplication as they are independent events in probability. While normal intuition would inform that the individual with 50% likelihood is the one that should be overfilled, in reality, it is the orange and purple Shift Groups that should be overfilled due to their respective 56% and 84% chances of a cancellation occurring.

When overfilling a Shift Group becomes necessary, SF will generate a shift that matches all of the same data as in the Shift Group (this will be called a ghost shift). This ghost shift can be booked and claimed, but does not represent a real shift. HCPs are not given information that they have signed up for a ghost shift. If any shift in the Shift Group is cancelled, this ghost shift is turned into a real shift (comes alive).

7. **What about if a Shift Group is overfilled and every HCP shows up?**

In cases where a Shift Group is overfilled and all initially scheduled healthcare professionals (HCPs) fulfill their shifts as planned, SF will employ a cancellation process for the ghost shift on a first-come, first-served basis. While assessing the quality of HCPs would be an optimal solution, introducing such complexity during the initial rollout is unwarranted, particularly as there is no existing data available to evaluate HCP quality reliably.

8. **When do we decide to cancel a shift? Who is paying for it?**

SF will cancel the shifts with an advanced notice of 4 hours, as this prevents situations where a HCP is underway to a HCF only to receive a cancellation as they arrive. Additionally, this covers most shifts that are cancelled (with a positive lead time), approximately 85%.

SF will pay the cost for cancellation of ghost shifts. If a ghost shift is cancelled, the HCP will be paid 2 hours of pay. A fast back of the napkin calculation shows that 99.95% (this calculation includes 0 hour shifts) of shifts are 2 hours or greater, thus we appropriately compensate the majority of HCPs. Additionally, this reduces the amount of HCPs that are overcompensated relative to their actual claimed shifts.

9. **Will this be a profitable operation?**

Based on 5000 additional filled shifts through overfilling, SF expects $348,334.16 in extra revenue. The cost to compensate for non-deleted and non-negative/zero hour shift groups (3812) is approximately $315,774.79, resulting in a profit ratio of $1.1. These calculations do not include the potential revenue from reduced churn rates of healthcare facilities (HCFs), with each HCF estimated to be worth around $17,000 in revenue to SF. Eliminating churn from the top 5 highest-value HCFs alone represents $74,000 in revenue. Overall, the overfilling operation is expected to be profitable for SF.

10. **Is this the best customer experience you can think of, unconstrained? If not why not? What constraints (internal, external) prevent you from your best customer experience idea?**

No, this is not the best customer experience absolutely possible. An ideal situation is twofold:

- For HCFs: 100% of posted shifts remain filled with the highest quality HCPs
- For HCPs: Whenever a shift is signed up for, they are guaranteed to have that shift or if the shift is cancelled, they are offered a similar one in the same geographic region.

The primary constraint that realistically prevents the best customer experience situation from occurring is simple supply. In many markets, there are not enough quality workers that can be allocated to every shift posted on SF. In markets where there are overabundance of workers, there is the issue where there is not enough shifts to be distributed to all HCPs.

11. **Premortem: what's the most likely cause of failure if this fails?**

The most likely cause of failure is limited user acceptance or engagement. The adoption and acceptance of the overfilling function by both healthcare facilities and professionals are crucial for its success. If users do not embrace the concept, experience difficulties in using the function, or perceive it as burdensome, they may be less inclined to participate, limiting the overall effectiveness of the overfilling approach. To alleviate these concerns, it is vital that HCPs receive notifications so that they do not appear to ghost shifts. It is also important to not have excessive ghost shift cancellations, as it will lead to an inverse situation where HCPs will choose to not work with specific HCFs.

**12. Rollout plan? What if it goes badly during rollout and what will we do?**

The current rollout plan is as follows (a sprint represents 2 weeks):

Phase 1: Initial Design (1 sprint)
Phase 2: Pre-Rollout Preparation (6-7 sprints)

- Design for capturing Shift Group information, creating ghost shifts, merge ghost shifts, and eliminating ghost shifts (2-3 sprints)
- Develop and implement necessary backend changes and APIs to support overfilling (1 sprint)
- Remove logic for billing in ghost shifts (0.5 sprint)
- Hiding ghost shifts from HCF (0.5 sprint)
- Notifications for ghost shifts (1 sprint)
- Perform unit testing and integration testing to ensure the stability and functionality of the new system (1 sprint)

Phase 2: Soft Launch (1 sprint)

- Release the overfilling function to test market for initial testing (1 sprint)
    - Monitor change in book rates of ghost and real shifts. Both should see an increase.
    - Monitor change in actual worked shifts. It is key to see an increase here. If there is not an increase noticed, modify probability threshold required for overfilling by 5%.
    - Monitor change in ghost shift generation. If there is a substantial increase in ghost shift generation (over 50% spike), cease operation here and re-evaluate criteria for ghost shift generation. There should not be such a substantial spike.

Phase 3: Go/No-Go Decision

- On results of soft launch, did the overfilling function see an increase in actual worked shifts (percentage)? If not, rework criteria and re-launch in different test market.
- If there are multiple failing soft launches, pull out overfilling function and examine potential design flaws or flawed solutions.

Phase 4: Full Rollout (2 sprints) (Post Go Decision)

- Gradually expand the availability of the overfilling function to a other test markets (1 sprint)
- Monitor system performance and user adoption rates during the rollout (1 sprint)
- Address any technical or performance issues that arise during the rollout (ongoing)
- Collect user feedback and make iterative enhancements based on user input (ongoing)

Phase 5: Post-Rollout Evaluation and Optimization (1-2 sprints)

- Conduct a comprehensive evaluation of the overfilling function's effectiveness and impact (0.5 sprint)
- Analyze user adoption rates, satisfaction surveys, and key performance indicators (0.5 sprint)
- Identify areas for improvement and prioritize future enhancements (0.5 sprint)
- Implement necessary optimizations and updates based on the evaluation (0.5 sprint)

13. **What's the plan for training internal teams and external customers?**

External customers will not be educated regarding the usage of ghost shifts, as they should not have visibility on that function.

For internal support personnel, there will be documentation provided and initial response templates regarding what a ghost shift is and how to respond to customer concerns. For HCPs that experience errors where they arrive at a mismatched ghost shift, they will be instructed to inform the HCP that backup shifts are generated on behalf of HCFs and they will be compensated.

For HCFs, they will be informed that a separate individual has been found for their shift via both SMS and email. The new individual's documentation will be made available on the app.

**Customer FAQs**

**HCF FAQ**

1. **Who is this person who showed up to my facility? They claim they were sent by SF.**

SF generates backup shifts to prevent situations where healthcare professionals do not show up to their assigned shifts. There may be occasional issues where the backup shift is not properly canceled, and the backup healthcare professional is sent in. You may send this individual home and SF will compensate them accordingly.

2. **Is there any way to see the backup shifts?**

Presently, there is no way to see the backup shifts associated with your facility. We perform this action on our backend to ensure that it is as smooth and seamless as possible for you.

3. **We don't want this function for our facility, can we turn it off?**

No, this function may not be turned off. This function is designed for the benefit of our customers to reduce disruptions caused by no-show, no-call individuals.

4. **Will we be charged for these backup shifts?**

No, SF takes full financial responsibility for the backup shifts and compensates the healthcare professionals accordingly for any issues that may arise.

**HCP FAQ**

1. **My shifts are getting cancelled 4 hours before they start, what's happening?**

SF Heath utilizes an overfilling system to ensure that every shift is fulfilled to its fullest capacity. For any shift that is cancelled as a result of this function, you will be compensated for 2 hours of work.

2. **Is this going to be a common occurrence?**

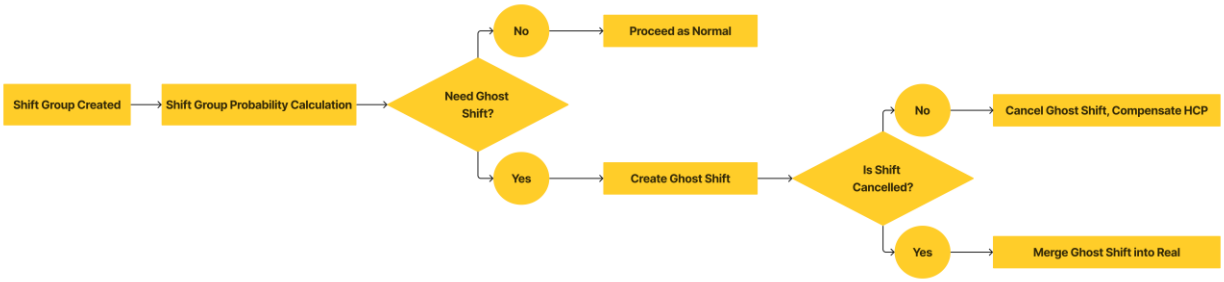
Overfilling is implemented as a proactive measure to ensure sufficient shift coverage, but its occurrence may vary depending on factors such as the demand for shifts and the availability of

healthcare providers. It is intended to be a tool that helps both healthcare professionals and facilities.

3. **I showed up to a shift but the facility said the shift never existed in the first place?**

SF uses an overfilling function to ensure that every shift is fulfilled. Occasionally, there are situations where an error occurs, and you may not be appropriately notified about a shift cancellation or it does not occur in the first place. In this instance, you will receive 2 hours of compensation for your time.

## Appendix

### Design Mockups

The below displays a preliminary diagram for perceived functionality of overfilling function.